

4. Velocity of a rigid body.

4.1. Rotational velocity.

pure rotation $R_{ab}(t) \in SO(3)$: Trajectory of frame B from the origin of frame A.

(A : spatial frame, B : body frame)

g_b : fixed pt in frame B $\Rightarrow g_a(t) = R_{ab}(t) g_b$

$$v_a(t) = \dot{g}_a(t) = \dot{R}_{ab}(t) g_b = \dot{R}_{ab}(t) \underbrace{R_{ab}(t)^T}_{g_a(t)} \underbrace{R_{ab}(t) g_b}_{g_a(t)} = \dot{R}_{ab}(t) R_{ab}(t)^T g_a(t)$$

Lemma 2.12. For $R(t) \in SO(3)$, $\dot{R}(t) R^T(t) \in so(3)$, $R^T(t) \dot{R}(t) \in so(3)$ (skew-symmetric).

pf) $R(t) R(t)^T = I \Rightarrow \dot{R} R^T + R \dot{R}^T = 0$

$$\Rightarrow \dot{R} R^T = -\dot{R} R^T = -R \dot{R}^T = -(\dot{R} R^T)^T = -(R \dot{R}^T)^T$$

$$R(t)^T R(t) = I \Rightarrow \dots \Rightarrow R^T(t) \dot{R}(t) \in so(3)$$

$\dot{R}_{ab}(t) R_{ab}(t)^T =: \hat{\omega}_{ab}^s \in so(3)$, $\omega_{ab}^s \in \mathbb{R}^3$: instantaneous spatial angular velocity.

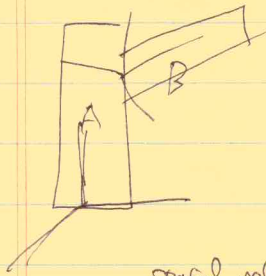
$$(v_{g_a}(t) = \hat{\omega}_{ab}^s g_a(t) = \omega_{ab}^s \times g_a(t))$$

~~$\dot{R}_{ab}(t)^T R_{ab}(t) =: \hat{\omega}_{ab}^b \in so(3)$, $\omega_{ab}^b \in \mathbb{R}^3$: instantaneous body angular velocity.~~

~~$$\begin{aligned} (g_a: \text{fixed pt in frame A} \Rightarrow g_a(t) = R_{ab}(t)^T g_a &\Rightarrow v_{g_a}(t) = \dot{R}_{ab}(t)^T g_a \\ &= \dot{R}_{ab}(t)^T R_{ab}(t) R_{ab}(t)^T g_a = \hat{\omega}_{ab}^b g_a = \omega_{ab}^b \times g_a) \end{aligned}$$~~

$$(v_{g_b}(t) = R_{ab}(t)^T v_{g_a}(t) = R_{ab}(t)^T \hat{\omega}_{ab}^b g_b = \hat{\omega}_{ab}^b g_b = \omega_{ab}^b \times g_b)$$

4.2. Rigid body velocity



$$g_{ab}(t) = \begin{pmatrix} R_{ab}(t) & p_{ab}(t) \\ 0 & 1 \end{pmatrix} \in SE(3)$$

spatial velocity $\hat{V}_{ab}^s = \dot{g}_{ab} g_{ab}^{-1} = \begin{pmatrix} \dot{R}_{ab} & \dot{p}_{ab} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} R_{ab}^T & -R_{ab}^T p_{ab} \\ 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} \dot{R}_{ab} R_{ab}^T & -\dot{R}_{ab} R_{ab}^T p_{ab} + \dot{p}_{ab} \\ 0 & 0 \end{pmatrix}$$

$$V_{ab}^s = \begin{pmatrix} -\dot{R}_{ab} R_{ab}^T p_{ab} \\ (\dot{R}_{ab} R_{ab}^T)^v \end{pmatrix} = \begin{pmatrix} V_{ab}^s \\ \omega_{ab}^s \end{pmatrix}$$

* $\omega \in \mathbb{R}^3 \mapsto \hat{\omega} \in \mathfrak{so}(3)$
 $(\hat{\omega})^v = \omega \leftarrow$

$$\dot{r}_a = \dot{g}_{ab} r_b \Rightarrow V_{ra} = \dot{g}_{ab} r_b = \boxed{\dot{g}_{ab} g_{ab}^{-1}} r_a = \hat{V}_{ab}^s r_a$$

$$\Rightarrow V_{ra} = \omega_{ab}^s \times r_a + V_{ab}^s$$

body velocity $\hat{V}_{ab}^b = g_{ab}^{-1} \dot{g}_{ab} = \begin{pmatrix} R_{ab}^T & -R_{ab}^T \dot{p}_{ab} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \dot{R}_{ab} & \dot{p}_{ab} \\ 0 & 0 \end{pmatrix}$

$$= \begin{pmatrix} R_{ab}^T \dot{R}_{ab} & R_{ab}^T \dot{p}_{ab} \\ 0 & 0 \end{pmatrix}$$

$$V_{ab}^b = \begin{pmatrix} V_{ab}^b \\ \omega_{ab}^b \end{pmatrix} = \begin{pmatrix} R_{ab}^T \dot{p}_{ab} \\ (R_{ab}^T \dot{R}_{ab})^v \end{pmatrix}$$

$$V_{rb} = g_{ab}^{-1} V_{ra} = g_{ab}^{-1} \dot{g}_{ab} r_b = \hat{V}_{ab}^b r_b$$

$$\Rightarrow V_{rb} = \omega_{ab}^b \times r_b + V_{ab}^b$$

relation between spatial and body velocity.

$$\begin{aligned}\hat{v}_{ab}^s &= g_{ab} g_{ab}^T = g_{ab} (g_{ab}^T g_{ab}) g_{ab}^{-1} = g_{ab} \hat{v}_{ab}^b g_{ab}^{-1} \\ &= \begin{pmatrix} R_{ab} & p_{ab} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{w}_{ab}^b & v_{ab}^b \\ 0 & 0 \end{pmatrix} \begin{pmatrix} R_{ab}^T & -R_{ab}^T p_{ab} \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} R_{ab} \hat{w}_{ab}^b R_{ab}^T & -R_{ab} \hat{w}_{ab}^b R_{ab}^T p_{ab} + R_{ab} v_{ab}^b \\ 0 & 0 \end{pmatrix}\end{aligned}$$

$$\Rightarrow \begin{cases} \hat{w}_{ab}^s = R_{ab} \hat{w}_{ab}^b R_{ab}^T = (R_{ab} \hat{w}_{ab}^b)^T \\ v_{ab}^s = -R_{ab} \hat{w}_{ab}^b R_{ab}^T p_{ab} + R_{ab} v_{ab}^b = -\hat{w}_{ab}^s p_{ab} + R_{ab} v_{ab}^b \\ = -(R_{ab} \hat{w}_{ab}^b)^T p_{ab} + R_{ab} v_{ab}^b \end{cases}$$

$$\Rightarrow \begin{cases} w_{ab}^s = R_{ab} w_{ab}^b \\ v_{ab}^s = \cancel{w_{ab}^s p_{ab}} - \hat{w}_{ab}^s p_{ab} + R_{ab} v_{ab}^b \\ = -(R_{ab} \hat{w}_{ab}^b)^T p_{ab} + R_{ab} v_{ab}^b = -(\hat{p}_{ab}^b w_{ab}^b) \times p_{ab} + R_{ab} v_{ab}^b \\ = p_{ab} \times (R_{ab} \hat{w}_{ab}^b) + \dots = \hat{p}_{ab}^b w_{ab}^b + \dots \end{cases}$$

$$\hat{v}_{ab}^s = \begin{pmatrix} v_{ab}^s \\ w_{ab}^s \end{pmatrix} = \underbrace{\begin{pmatrix} R_{ab} & p_{ab} R_{ab}^T \\ 0 & R_{ab} \end{pmatrix}}_{Ad_{g_{ab}}} \begin{pmatrix} v_{ab}^b \\ w_{ab}^b \end{pmatrix} = \hat{p}_{ab}^b w_{ab}^b + \dots$$

$$\hat{v}_{ab}^s = \begin{pmatrix} v_{ab}^s \\ w_{ab}^s \end{pmatrix} = \begin{pmatrix} R_{ab} & p_{ab} R_{ab}^T \\ 0 & R_{ab} \end{pmatrix} \begin{pmatrix} v_{ab}^b \\ w_{ab}^b \end{pmatrix}$$

$g \in SE(3) \Rightarrow Ad_g : \mathfrak{se}(3) \rightarrow \mathfrak{se}(3)$ adjoint transformation.

$$Ad_g = \begin{pmatrix} R & \hat{p}R \\ 0 & R \end{pmatrix}$$

body velocity $\rightarrow$ spatial velocity

lem 2.13 $\xi \in \mathfrak{se}(3)$, $g \in SE(3) \Rightarrow g \xi g^{-1}$ twist with twist coordinate $Ad_g \xi \in \mathfrak{se}(3)$

pf) same as before

$$g = \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix} \quad \xi = \begin{pmatrix} \hat{w} & v \\ 0 & 0 \end{pmatrix}$$

$$g \xi g^{-1} = \begin{pmatrix} R & \hat{p}R \\ 0 & R \end{pmatrix} \begin{pmatrix} \hat{w} & v \\ 0 & 0 \end{pmatrix}$$

$$g \xi g^{-1} = (Ad_g \xi)^{\wedge}$$

4.4. Coordinate transformations

$$g_{ab}, g_{bc} \in SE(3) \Rightarrow g_{ac} = ?$$

Prop 2.14. $V_{ac}^s = V_{ab}^s + \text{Ad}_{g_{ab}} V_{bc}^s$

pf) $\hat{V}_{ac}^s = \hat{g}_{ac} \hat{g}_{ac}^{-1} = \underset{g}{(g_{ab}^0 g_{bc}^0 + g_{ab} g_{bc}^0)} (\hat{g}_{bc}^{-1} \hat{g}_{ab}^{-1})$
 $g_{ac} = g_{ab} g_{bc}$

$$= g_{ab}^0 g_{ab}^{-1} + g_{ab} (g_{bc}^0 g_{bc}^{-1}) g_{ab}^{-1}$$

$$= \hat{V}_{ab}^s + \underbrace{g_{ab} \hat{V}_{bc}^s g_{ab}^{-1}}_{\text{II}} = (\text{Ad}_{g_{ab}} V_{bc}^s)^{\wedge}$$

$$\Rightarrow V_{ac}^s = V_{ab}^s + \text{Ad}_{g_{ab}} V_{bc}^s$$

Prop 2.15. $V_{ac}^b = \text{Ad}_{g_{bc}^{-1}} V_{ab}^b + V_{bc}^b$

pf, same way.

Lemma 2.16. $V_{ab}^b = -V_{ba}^s$

$$V_{ab}^b = -\text{Ad}_{g_{ba}} V_{ba}^b$$

pf) $\hat{V}_{ab}^b = \hat{g}_{ab}^{-1} \hat{g}_{ab}^0$ II $\hat{V}_{ba}^s = \hat{g}_{ba}^{-1} \hat{g}_{ba}^0$

$$= \hat{g}_{ab} (g_{ab}^T g_{ab}^0 g_{ab}^{-1})$$

$$= g_{ab}^0 g_{ab}^T = -\hat{V}_{ab}^b$$

$$g g^{-1} = I$$

↓

$$\dot{g} g^{-1} + g (\dot{g}^{-1}) = 0$$

⇓

$$(\dot{g}^{-1}) = -g^{-1} \dot{g} g^{-1}$$

$$V_{ba}^s = \text{Ad}_{g_{ba}} V_{ba}^b$$

7. Wrenches and Reciprocal Screws

9.1. Wrenches

$$F = \begin{pmatrix} f \\ \tau \end{pmatrix} \quad \begin{array}{l} \text{generalized force} \\ \text{wrench} \end{array} \quad \begin{array}{l} f \in \mathbb{R}^3 : \text{linear component, force} \\ \tau \in \mathbb{R}^3 : \text{rotational component, torque} \end{array}$$

F_b : wrench applied at the origin of B

motion of rigid body $g_{ab}(t)$, A : inertial frame, B : frame attached to the rigid

V_{ab}^b : body velocity

$$\Rightarrow \dot{W} = V_{ab}^b \cdot F_b = v \cdot f + \omega \cdot \tau$$

instantaneous work

$$\text{total work over } [t_1, t_2] = \int_{t_1}^{t_2} V_{ab}^b \cdot F_b dt$$

- Two wrenches are equivalent if they generate the same work

F_b applied at the origin of B .

Find F_c applied at C which is equivalent to F_b .

g_{bc} : frame C relative to B

$$V_{ac}^b \cdot F_c = V_{ab}^b \cdot F_b = (Ad_{g_{bc}} V_{ac}^b)^T F_b = V_{ac}^b \cdot Ad_{g_{bc}}^T F_b$$

$$\Rightarrow F_c = Ad_{g_{bc}}^T F_b$$

If several wrenches applied to a rigid body, we express the wrenches in one same frame, and sum up the wrench vectors to get a net wrench.

(ex) multifingered grasp. F_{c_i} : wrench on the grasped object by i -th finger.

$$\text{net wrench in the frame } O : F_o = \sum_i Ad_{g_{oc_i}}^T F_{c_i}$$

Chasles's thm every twist is generated by a screw.

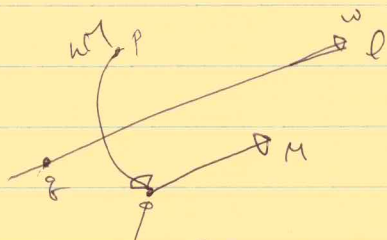
Poincaré's thm every wrench is equivalent to a force and torque ^{applied} along the same axis.

Wrench along a screw S .

S : screw with axis $\ell = \{q + \lambda w : \lambda \in \mathbb{R}\}$, $\|w\|=1$, pitch h , magnitude M .

wrench defined by applying $\left\{ \begin{array}{l} \text{a force of magnitude } M \text{ along } \ell \\ \text{a torque of magnitude } hM \text{ about } \ell. \end{array} \right.$

$h = \infty$: pure ~~force~~ w/ magnitude M
torque



$$F = M \begin{pmatrix} w \\ -w \times q + hw \end{pmatrix}, \quad h \text{ finite}$$

$$F = M \begin{pmatrix} 0 \\ w \end{pmatrix}, \quad h = \infty$$

$$\frac{hMw \times (p-q)}{\|hMw \times (p-q)\|}$$

$$hM(-w \times q)$$

$$(-w \times q + hw) \times p$$

$$-(w \times q_p) \times p$$

$$\tau = \underbrace{\tau_{\text{force}}}_{q \times Mw} + \underbrace{\tau_{\text{torque}}}_{hMw}$$

F : wrench along S .

Poincaré

every wrench applied to a rigid body is equivalent to a force and torque applied along a fixed axis.

pf) $F = (f, \tau)$: net wrench

$F = 0$: trivial

$f = 0$ (pure torque): set $M = \|\tau\|$, $w = \tau / M$, $h = \infty$.

$f \neq 0$: $M = \|f\|$, $w = f / M$.

$$\text{solve } M(q \times w + hw) = \tau \quad \text{in } q, h$$

$$h = \frac{f^T \tau}{\|f\|^2}, \quad q = \frac{f \times \tau}{\|f\|^2} \Rightarrow M(q \times w + hw) = \tau$$

parallel to w perp to w .

screw coordinates of a wrench $F = (f, c)$

1. Pitch : $h = \frac{f \tau c}{\|f\|^2} \quad (f=0 \rightarrow h=\infty)$

2. Axis : $l = \begin{cases} \left\{ \frac{f \times c}{\|f\|^2} + \lambda f : \lambda \in \mathbb{R} \right\} & f \neq 0 \\ \{ 0 + \lambda c : \lambda \in \mathbb{R} \} & f = 0 \end{cases}$

3. Magnitude : $M = \begin{cases} \|f\| & f \neq 0 \\ \|c\| & f = 0 \end{cases}$

twist $\xleftarrow{\text{dual}} \text{wrench}$

$h=0$: pure rotation

pure force

$h=\infty$: pure translation

pure torque

3.3. Reciprocal screws

dot product of twist and wrench $\rightarrow$ instantaneous power

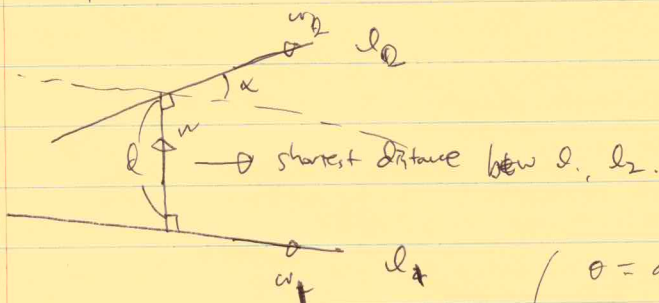
- wrench F is reciprocal to twist $V \Leftrightarrow F \cdot V = 0$

Def 2.3. Reciprocal screws

screws S_1 and S_2 are reciprocal if twist V about S_1 and wrench F along S_2 are reciprocal.

- reciprocal product

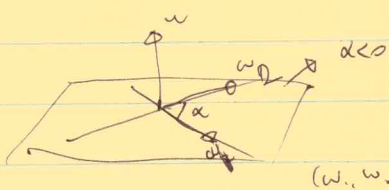
S_i : screw with $(l_i = \{p_i + \lambda w_i\}, h_i, M_i)$



α : angle between w_1 & w_2

$$\alpha = \arctan 2(w_1 \times w_2 \cdot n, w_1 \cdot w_2)$$

$$\theta = \arctan 2(y, x) \Leftrightarrow (\alpha \theta, s, \theta) = \frac{(x, y)}{\sqrt{x^2 + y^2}}$$



(w_1, w_2, w)

θ : o-oriented angle from w_1 to w_2

$$\theta \in (-\pi, \pi]$$

reciprocal product

$$S_1 \odot S_2 = \begin{cases} M_1 M_2 ((h_1 + h_2) \cos \alpha - d \sin \alpha) & \begin{cases} h_1 = \infty, M_1 \rightarrow 0 & M_1 M_2 \cos \alpha \\ h_2 = \infty & M_1 M_2 \cos \alpha \\ h_1 = 0, h_2 = 0 & 0 \end{cases} \end{cases}$$

Prop 2.18. Two screws S_1 & S_2 are reciprocal $\Leftrightarrow S_1 \odot S_2 = 0$.

pf) $\odot$ h_1, h_2 : finite

$$V : \text{twist about } S_1 \Rightarrow V = M_1 \begin{pmatrix} g_1 \times \omega_1 + h_1 \omega_1 \\ \omega_1 \end{pmatrix}$$

$$F : \text{wrench along } S_2 \Rightarrow F = M_2 \begin{pmatrix} \omega_2 \\ g_2 \times \omega_2 + h_2 \omega_2 \end{pmatrix}$$

where g_1, g_2 : points at $\underbrace{\text{axes}}_{\text{which}}$ are direct.

$$\Rightarrow g_2 = g_1 + d n$$

$$\begin{aligned} V \cdot F &= M_1 M_2 (\omega_2 \cdot (g_1 \times \omega_1 + h_1 \omega_1) + \omega_1 \cdot (g_2 \times \omega_2 + h_2 \omega_2)) \\ &= M_1 M_2 (\omega_2 \cdot g_1 \times \omega_1 + h_1 \omega_1 \cdot \omega_2 + \omega_1 \cdot (g_1 + d n) \times \omega_2 + h_2 \omega_1 \cdot \omega_2) \\ &= M_1 M_2 ((h_1 + h_2) \cos \alpha - d \sin \alpha) \\ &= S_1 \odot S_2. \end{aligned}$$

② $h_1 = \infty, h_2$: finite.

$$V = M_1 \begin{pmatrix} \omega_1 \\ 0 \end{pmatrix}, \quad F = M_2 \begin{pmatrix} \omega_2 \\ h_2 \omega_2 + h_2 \omega_2 \end{pmatrix}$$

$$= V \cdot F = M_1 M_2 \omega_1 \cdot \omega_2$$

$$V \cdot F = 0 \Leftrightarrow \omega_1 \perp \omega_2 \Leftrightarrow \alpha = \pm \frac{\pi}{2}$$

$$\textcircled{3} h_1 = 0, h_2 = \infty : \quad V = M_1 \begin{pmatrix} g_1 \times \omega_1 + h_1 \omega_1 \\ \omega_1 \end{pmatrix}, \quad F = M_2 \begin{pmatrix} 0 \\ \omega_2 \end{pmatrix}$$

$$V \cdot F = h_2 \omega_1 \cdot \omega_2$$

$$\textcircled{4} h_1 = \infty, h_2 = 0 : \quad V \cdot F = 0, \quad S_1 \odot S_2 = 0$$

- screws using twist coordinates $V_i = \begin{pmatrix} v_i \\ \omega_i \end{pmatrix}, \begin{pmatrix} v_i \\ \omega_i \end{pmatrix}$

$$\text{reciprocal product} = V_1 \odot V_2 = v_1^T \omega_2 + \omega_2^T v_1$$

$$\begin{aligned} v_i &= -\omega_i \times g_i + h_i \omega_i \\ &= (-\omega_1 \times g_1 + h_1 \omega_1)^T \omega_2 + (-\omega_2 \times g_2 + h_2 \omega_2)^T \omega_1 \\ &= -(\omega_1 \times g_1) \cdot \omega_2 - (\omega_2 \times g_2) \cdot \omega_1 + (h_1 + h_2) \omega_1 \cdot \omega_2 \\ &= (g_1 - g_2) \cdot (\omega_1 \times \omega_2) + (h_1 + h_2) \omega_1 \cdot \omega_2 \\ &= (h_1 + h_2) \cos \alpha - d \sin \alpha \end{aligned}$$

wrenches applied to an object
~~Reciprocal screws~~ ^{rigid body} $\rightarrow$ constraining screws.
 instantaneous (twists) that do not violate the constraints.

$\{s_1, \dots, s_k\}$: set of screws representing wrenches
 s_i : ~~screw~~ screw representing a twist.

$\rightarrow$ motion along s_i causes no work against $s_1, \dots, s_k$.

screws $\{s_1, \dots, s_k\} \rightarrow$ twists $\rightarrow$ form a linear subspace in $\mathbb{R}^6$.

S : reciprocal to $\{s_1, \dots, s_k\} \Rightarrow S$: reciprocal to any linear combination of
 $\{s_1, \dots, s_k\}$

$\{s_1, \dots, s_k\}$: system of screws.

$\{S: \text{reciprocal to } s_1, \dots, s_k\}$: reciprocal ~~system~~ screw system $\rightarrow$ linear
 algebra

Prop 2.19 r : dimension of system of screws $\{s_1, \dots, s_k\}$

$(s_i \rightarrow \text{twists / wrenches in } \mathbb{R}^6 \rightarrow \text{linear subspace in } \mathbb{R}^6)$

u : dimension of the corresponding reciprocal system.

$$\Rightarrow r + u = 6.$$